

Proof Toolkit

Implication: $P \Rightarrow Q$ is equivalent to contrapositive $\neg Q \Rightarrow \neg P$, not converse $Q \Rightarrow P$. For $P \Rightarrow Q$, prove both directions.

Direct proof: Unpack definitions; introduce arbitrary object; chain implications until target.

Contradiction: Assume target false and derive impossible statement, parity conflict, degree conflict, or violated extremal choice.

Induction: State $P(n)$; prove enough base cases; assume previous case(s); use recurrence/decomposition to prove next. Strong induction when proof needs many earlier cases.

Bijection: Define $f : X \rightarrow Y$; prove injective and surjective, or construct inverse $g : Y \rightarrow X$.

Double count: Choose one finite set of objects; count directly and by cases. If there is a sum, the index usually classifies by size, largest object, last step, component, or color.

Extremal method: Choose longest/shortest/largest/smallest object; show failure produces a more extreme object. Useful for paths, planar bounds, MST exchange, graph contradictions.

Invariant: Find quantity preserved or forced: parity, degree sum, component count, complement edge count, settled Dijkstra distance.

Exact value: Usually prove lower bound + upper bound. Construction gives lower/upper depending on minimization/maximization.

Common templates: Bijection: define/invert. Combinatorial identity: count same set two ways. Pigeonhole: choose boxes so avoiding target gives max capacity. IE: define bad sets. Graph non-iso: compare invariants. Bipartite: partition/2-color/no odd cycle. Euler: connected+degree parity. Hamilton: explicit order or obstruction. Coloring: coloring+lower bound. Tree: pick best equivalence.

Counting Principles

Objects and core rules

Encodings: Set = unordered/no repeats; sequence/string = ordered/repeats may occur; function $f : X \rightarrow Y =$ each x chooses exactly one y . Subset $A \subseteq [n] \Leftrightarrow$ binary characteristic string; hence $|\mathcal{P}(S)| = 2^{|S|}$. Labelled simple graph on n vertices $\Leftrightarrow$ subset of $\binom{n}{2}$ possible edges.

Product rule: Successive stages with $n_1, \dots, n_k$ choices give $n_1 \cdots n_k$. Use only when every earlier choice leaves stated number of later choices.

Sum rule: If $A_1, \dots, A_k$ are pairwise disjoint, then $|A_1 \cup \dots \cup A_k| = \sum_i |A_i|$. Split into a genuine partition before adding.

Bijection/division: $X \cong Y \Rightarrow |X| = |Y|$. If each final object represented exactly k times, final count = $|X|/k$.

Complement: Number good = $|U|$ - number bad. Always define the universe U first.

Inclusion-exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$; $|A \cup B \cup C| = |A| + |B| + |C| - |AB| - |AC| - |BC| + |ABC|$; in general $|\bigcup_{i=1}^n A_i| = \sum |A_i| - \sum |A_i A_j| + \sum |A_i A_j A_k| - \dots + (-1)^{n-1} |A_1 \cdots A_n|$.

IE usage: Use for overlapping bad events: surjections, upper bounds, divisibility, forbidden positions/strings. Element in exactly r sets contributes $\binom{r}{1} - \binom{r}{2} + \dots + (-1)^{r-1} \binom{r}{r} = 1$.

Pigeonhole and basic counts

PHP: If N objects enter k boxes, some box has at least $\lceil N/k \rceil$. Equiv.: if $N > rk$, some box has at least $r + 1$.

Common boxes: Residues mod m , parity classes, complementary pairs, graph degree values, geometric cells, sorted gaps.

Degree PHP: In a simple graph on $n \geq 2$ vertices, two vertices have same degree because possible degrees are $0, 1, \dots, n-1$, but 0 and $n-1$ cannot coexist.

Ramsey $R(3, 3) = 6$: Fix a vertex. Among 5 incident edges, 3 same color; the 3 endpoints either contain same-color edge or give opposite-color triangle.

Subsets/strings: Subsets of n -set: 2^n ; r -subsets: $\binom{n}{r}$; binary strings length n : 2^n ; alphabet size m : m^n ; even subsets = odd subsets = 2^{n-1} for $n \geq 1$.

Special selections: Choose r including k fixed: $\binom{n-k}{r-k}$; excluding k forbidden: $\binom{n-k}{r}$; at least one of k marked: $\binom{n}{r} - \binom{n-k}{r}$.

Lattice paths: From $(0, 0)$ to (a, b) with right/up steps: $\binom{a+b}{a} = \binom{a+b}{b}$.

Permutations & Combinations

Selections without repetition

Permutations: n distinct objects: $n!$. r -permutations: $P(n, r) = n(n-1) \cdots (n-r+1) = n!/(n-r)!$.

Combinations: r -combinations: $\binom{n}{r} = n!/[r!(n-r)!]$. Choose then order: $P(n, r) = \binom{n}{r} r!$. Symmetry: $\binom{n}{r} = \binom{n}{n-r}$.

Pascal / hockey: $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$; $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$; equivalently $\sum_{i=0}^n \binom{i+k}{k} = \binom{n+k+1}{k+1}$.

Binomial sums: $\sum_{k=0}^n \binom{n}{k} = 2^n$; $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$ for $n \geq 1$; $\sum k \binom{n}{k} = n2^{n-1}$; $\sum k(k-1) \binom{n}{k} = n(n-1)2^{n-2}$; $\sum k^2 \binom{n}{k} = n(n+1)2^{n-2}$.

Vandermonde: $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$. Special: $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.

Circular, blocks, gaps

Circular: Circular arrangements of n distinct objects: $(n-1)!$. If reflections also identified: $(n-1)!/2$ for $n \geq 3$. Fix one object to break rotation.

Choose around circle: Choose r from n , arrange around cycle: $P(n, r)/r = n!/[r(n-r)!]$.

Blocks: Required together: merge into one block, arrange block internally. Required apart: total minus together, or use gap method. If block placements overlap, refine by positions or use IE.

Nonadjacent positions: Line: choose r nonadjacent positions from n : $\binom{n-r+1}{r}$. Circle: $\frac{n}{n-r} \binom{n-r}{r}$ when valid.

Burnside: Group action count: number of orbits = $\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|$. Ordinary division by $|G|$ valid only when every orbit has size $|G|$.

Repetition, expansions, stars and bars

Repeated objects: Ordered selections with repetition: n^r . Unordered selections with repetition: $\binom{n+r-1}{r} = \binom{n+r-1}{n-1}$.

Multiset permutations: If multiplicities $n_1 + \dots + n_k = n$, count = $n!/(n_1! \cdots n_k!)$. Distinct objects into distinct boxes of sizes n_i : same formula. Identical objects into indistinguishable boxes: integer partitions, not stars-and-bars.

Multinomial: $(x_1 + \dots + x_m)^n = \sum_{n_1 + \dots + n_m = n} \frac{n!}{n_1! \cdots n_m!} x_1^{n_1} \cdots x_m^{n_m}$. Coefficient of $x_1^{n_1} \cdots x_m^{n_m}$ is $n!/(n_1! \cdots n_m!)$.

Binomial theorem: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$. Coeff. of $x^{n-k} y^k$ in $(ax+by)^n$ is $\binom{n}{k} a^{n-k} b^k$. Also $[x^r](1+x)^n = \binom{n}{r}$.

Stars and bars: $x_1 + \dots + x_k = N$, $x_i \geq 0$: $\binom{N+k-1}{k-1}$. If $x_i \geq \ell_i$, set $x_i = \ell_i + y_i$, count $\binom{N-\sum \ell_i + k-1}{k-1}$.

Inequality / upper bounds: $x_1 + \dots + x_k \leq N$: add slack $x_{k+1} \geq 0$, count $\binom{N+k}{k}$. For $0 \leq x_i \leq u_i$, use $\sum_{S \subseteq [k]} (-1)^{|S|} \binom{N-\sum_{i \in S} (u_i+1) + k-1}{k-1}$, treating impossible binomial terms as 0.

Monotone sequences: $1 \leq a_1 \leq \dots \leq a_r \leq n$: $\binom{n+r-1}{r}$. Strictly increasing $1 \leq a_1 < \dots < a_r \leq n$: $\binom{n}{r}$.

Generating functions: Allowed values of x_i become factors: $x_i \geq 0 \mapsto (1-x)^{-1}$; $0 \leq x_i \leq b \mapsto 1+x+\dots+x^b$; $x_i \geq a \mapsto x^a/(1-x)$. Count = $[x^N] \prod_i F_i(x)$.

Functions and Counting

All functions: Let $|X| = m$, $|Y| = n$. All functions $X \rightarrow Y$: n^m .

Injective: Possible iff $m \leq n$; count $P(n, m) = n!/(n-m)!$.

Bijjective: Possible iff $m = n$; count $n!$.

Surjective: Possible iff $m \geq n$; count $\# \text{Surj}(X, Y) = \sum_{j=0}^n (-1)^j \binom{n}{j} (n-j)^m$. IE interpretation: subtract functions missing at least one codomain element.

Stirling viewpoint: Distinct objects into j indistinguishable nonempty boxes: $S(n, j) = \frac{1}{j!} \sum_{i=0}^j (-1)^i \binom{j}{i} (j-i)^n$. Distinct objects into j distinguishable nonempty boxes: $j!S(n, j)$.

Recurrence Relations

Modelling and standard recurrences

Setup: Define a_n precisely; split by first/last step, last symbol, final tile, or state; prove cases disjoint/exhaustive; state valid n -range and initial values;

check small n .

State refinement: If one state loses information, introduce auxiliary states, derive coupled recurrences, then add/eliminate.

Hanoi: $H_1 = 1$, $H_n = 2H_{n-1} + 1$, so $H_n = 2^n - 1$.

No consecutive zeros: Binary strings: $a_1 = 2$, $a_2 = 3$, $a_n = a_{n-1} + a_{n-2}$ by endings 1 or 01. Ternary strings: $a_n = 2a_{n-1} + 2a_{n-2}$.

Tiling: Classify by last/first tile width. Empty tiling often $a_0 = 1$. Fibonacci tiling usually $a_n = a_{n-1} + a_{n-2}$.

Even zeros: Decimal strings with even number of zero digits: $a_0 = 1$, $a_n = 8a_{n-1} + 10^{n-1}$.

Linear homogeneous recurrences

Order 1: $a_n = Aa_{n-1}$ gives $a_n = a_0 A^n$.

Order 2: $a_n = Aa_{n-1} + Ba_{n-2}$, characteristic $r^2 - Ar - B = 0$. Distinct roots $r_1 \neq r_2$: $a_n = C_1 r_1^n + C_2 r_2^n$. Repeated root r : $a_n = (C_1 + C_2 n)r^n$.

General order: $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$, $c_k \neq 0$; characteristic $r^k - c_1 r^{k-1} - \dots - c_k = 0$.

Repeated roots: Root r of mult m contributes $(C_0 + \dots + C_{m-1} n^{m-1})r^n$.

Complex roots: Roots $\rho(\cos \theta \pm i \sin \theta)$ contribute $\rho^n (C_1 \cos n\theta + C_2 \sin n\theta)$.

Constants: Use initial conditions only after writing the general solution form. Need as many independent initial equations as constants.

Non-homogeneous recurrences

Structure: $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$. General solution $a_n = a_n^{(h)} + a_n^{(p)}$, where $a^{(h)}$ solves the associated homogeneous recurrence.

Affine first-order: $a_n = Aa_{n-1} + B$. If $A \neq 1$, $a_n = CA^n + B/(1-A)$; if $A = 1$, $a_n = a_0 + nB$.

Trial forms: Forcing $c \mapsto A$; polynomial $P_t(n) \mapsto Q_t(n)$; $s^n \mapsto As^n$; $P_t(n)s^n \mapsto Q_t(n)s^n$; $\cos \theta n, \sin \theta n \mapsto A \cos \theta n + B \sin \theta n$; sums $\mapsto$ sum trials

Generating functions: $A(x) = \sum_{n \geq 0} a_n x^n$; multiply recurrence by x^n , sum over valid n , solve for $A(x)$, extract coefficients.

Graph Theory Basics

Definitions, notation, standard graphs

Graph notation: $G = (V, E)$, $n = |V|$, $m = |E|$. Simple undirected: $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$. Adjacent: $uv \in E$. Neighbourhood $N_G(v) = \{u : uv \in E\}$, degree $\deg_G(v) = |N_G(v)|$.

Walk / trail / path: Walk: $v_0 e_1 v_1 \cdots e_k v_k$. Trail: no repeated edge. Path/simple path: no repeated vertex. Closed walk: $v_0 = v_k$. Circuit: closed trail. Cycle: closed path of length ≥ 3 .

Subgraph language: Subgraph $H = (W, F)$: $W \subseteq V$, $F \subseteq E$. Induced $G[W]$: all edges of G with both endpoints in W . Spanning subgraph: $W = V$.

Standard graph data / tutorial graphs.

Graph	$ V $	$ E $	degree / key facts
K_n	n	$\binom{n}{2}$	all deg $n-1$; $\chi = n$; bip iff $n \leq 2$; Ham cycle iff $n \geq 3$; Euler circuit iff n odd
K_4	4	6	planar 3-regular tetrahedral graph; also W_4
K_5	5	10	nonplanar since $10 > 3(5) - 6 = 9$; $K_5 - e$ planar with 9 edges
P_n	n	$n-1$	path graph; two leaves if $n \geq 2$, internal deg 2; bip; tree; Ham path yes
C_n	n	n	cycle graph; all deg 2; bip iff n even; $\chi = 2/3$ even/odd; Euler+Ham cycle yes
$K_{p,q}$	$p+q$	pq	p vertices deg q , q vertices deg p ; bip; tree iff $p = 1$ or $q = 1$
W_n	n	$2(n-1)$	wheel: hub deg $n-1$, rim deg 3; planar; regions $r = n$; $\chi = 3$ if n odd, 4 if even
Tree	n	$n-1$	connected acyclic; forest has $m = n - c$; at least two leaves if $n \geq 2$

$K_{m,n}$ degree sequence: $\underbrace{n, \dots, n}_m \text{ times}, \underbrace{m, \dots, m}_n \text{ times}$, then sort nonincreasing. Ex-

ample $K_{2,3} : 3, 3, 2, 2, 2$.

Counting graphs: Labelled simple graphs on fixed n vertices: $2^{\binom{n}{2}}$. With exactly r edges: $\binom{\binom{n}{2}}{r}$. Copies of K_s in K_n : $\binom{n}{s}$.

Degree facts: $\sum_{v \in V} \deg(v) = 2m$; odd-degree vertices occur in even num-

ber. k -regular $\Rightarrow nk = 2m$. Directed: $\sum \deg^+ = \sum \deg^- = m$.
Complement: $\overline{G} = (V, \binom{V}{2} \setminus E)$; $m + \bar{m} = \binom{n}{2}$; $\deg_{\overline{G}}(v) = n - 1 - \deg_G(v)$;
 $G \cong H \Rightarrow \overline{G} \cong \overline{H}$. Self-complementary $\Rightarrow n \equiv 0, 1 \pmod{4}$.

Isomorphism and graphical sequences

Isomorphism: $G \cong H$ iff $\exists$ bijection $f : V(G) \rightarrow V(H)$ such that $uv \in E(G) \Leftrightarrow f(u)f(v) \in E(H)$.
Non-isomorphism invariants: Compare n, m , degree sequence, components, isolated/pendant vertices, triangles, k -cycles, cut vertices, bridges, complements, neighbourhood structure. Same degree sequence $\nRightarrow$ isomorphic.
Constructing iso: Map unique degrees first; preserve adjacency and non-adjacency. Check $uv \in E(G) \iff f(u)f(v) \in E(H)$, not just edge count.
Havel–Hakimi: Graphical test: sort degree sequence descending; remove largest d ; subtract 1 from next d entries; resort. Graphical iff process reaches all zeros without negative/impossible entry. Even sum is necessary, not sufficient.

Connectivity / Traversal

Connectivity: G connected iff every pair of vertices has a path. Components = maximal connected subgraphs. Directed: strongly connected iff directed paths both ways; weakly connected iff underlying graph connected.
Component bounds: Acyclic graph with n vertices and c components has $m = n - c$. Minimum edges for c connected components: $n - c$. Maximum edges: $\binom{n-c+1}{2}$ by one large component plus $c - 1$ isolated vertices.
Euler vs Hamilton.

Feature	Euler	Hamilton
Uses	every edge exactly once	every vertex exactly once
Path count	edge m edges, $m + 1$ vertex appearances	$n - 1$ edges
Circuit/cycle count	m edges, closed	n edges, closed
Criterion	connected nonzero-edge part + odd-degree count 0 or 2	no simple criterion; use construction/obstruction
Circuit condition	all degrees even	explicit cycle through all vertices
Open points	end-precisely the two odd vertices	first/last vertices of listed order
Degree role	parity decisive	parity irrelevant; deg-1 forbids Hamilton cycle
Proof style	connectedness + parity	degree vertex ordering / cut obstruction

Euler criteria: Ignore isolated vertices for connectedness. Euler circuit $\iff$ nonzero-edge part connected and every vertex even. Euler path $\iff$ nonzero-edge part connected and exactly 0 or 2 odd vertices.
Hamilton proof: Path: exhibit $v_1, \dots, v_n$ with $v_i v_{i+1} \in E$. Cycle: additionally $v_n v_1 \in E$. A Hamilton path on n vertices has $n - 1$ edges; Hamilton cycle has n edges.
Hamilton obstructions: Bridge or cut vertex forbids Hamilton cycle. Degree-1 vertex forbids Hamilton cycle. Hamilton cycle $\Rightarrow c(G - S) \leq |S|$ for all nonempty S ; Hamilton path $\Rightarrow c(G - S) \leq |S| + 1$.
Bipartite Hamilton tests: If G has bipartition $A \sqcup B$, Hamilton cycle forces $|A| = |B|$; Hamilton path forces $||A| - |B|| \leq 1$. Thus $K_{m,n}$ has Hamilton cycle iff $m = n \geq 2$ and Hamilton path iff $|m - n| \leq 1$.
Longest paths: In a connected graph, any two longest simple paths intersect; otherwise a shortest connector creates a longer path.
Special graph tests.

Graph	Euler	Hamilton	bipartite / colour
K_n	circuit iff n odd; path iff $n \geq 1$	iff $n \geq 1$; cy-bip iff $n \leq 2$	$\chi = n$
P_n	path yes for $n \geq 2$; no circuit	Hamilton path yes; no Hamilton cycle	$\chi = 2$ if $n \geq 2$
C_n	Euler circuit yes	Hamilton cycle yes	bip iff n even; $\chi = 2/3$ even/odd
$K_{m,n}$	odd iff count 0 or 2	count cycle iff $m = n \geq 2$; path iff $ m - n \leq 1$	$\chi = 2$ if $mn > 0$
Tree	Euler path iff have odd degree	at Hamilton path iff graph P_n	$\chi = 2$ if non-trivial

Coloring / Planarity

Coloring and bipartite graphs

Coloring: k -coloring: function $c : V \rightarrow \{1, \dots, k\}$ with $uv \in E \Rightarrow c(u) \neq c(v)$. Chromatic number $\chi(G) = \min k$. Prove $\chi(G) = k$ by upper bound k -coloring + lower bound no $(k - 1)$ -coloring.
Coloring bounds: $\omega(G) \leq \chi(G) \leq \Delta(G) + 1$, where ω = clique number and Δ = max degree. If $K_r \subseteq G$, then $\chi(G) \geq r$. Deleting vertex: $\chi(G) - 1 \leq \chi(G - v) \leq \chi(G)$.
Bipartite definition: G bipartite iff $V = A \sqcup B$ and every edge has one endpoint in A and one in B , equivalently $E \subseteq \{ab : a \in A, b \in B\}$.
Bipartite criteria: G bipartite $\iff$ 2-colorable $\iff$ no odd cycle. Connected test: choose root r ; put vertices by parity of $\text{dist}(r, v)$; valid iff no edge joins same parity.
Prove bipartite: Either exhibit partition A, B and check all edges cross; or give a valid 2-coloring; or prove no odd cycle. For trees, use distance parity from root.
Disprove bipartite: Find an odd cycle. K_n bipartite iff $n \leq 2$. C_n bipartite iff n even.
Special χ values: $\chi(K_n) = n$; $\chi(C_n) = 2$ if n even, 3 if odd; $\chi(K_{p,q}) = 2$ for $p, q \geq 1$; trees with at least one edge have $\chi = 2$; K_4 is planar with $\chi = 4$.

Planarity

Euler formula: Connected planar graph: $n - m + r = 2$. With c components: $n - m + r = 1 + c$. Region-degree sum: $\sum_R \deg(R) = 2m$; bridges count twice on a face boundary.
Planar bounds: Connected simple planar, $n \geq 3$: $m \leq 3n - 6$. Triangle-free/bipartite planar: $m \leq 2n - 4$. If girth $g \geq 3$, then $m \leq \frac{g}{g-2}(n - 2)$.
Face-degree method: Use $\sum_R \deg(R) = 2m$. If every face has degree at least g , then $gr \leq 2m$; combine with Euler. Bridges count twice on a face.
No planar 6-regular: Simple planar $n \geq 3$ has average degree $2m/n \leq 6 - 12/n < 6$, so no simple planar 6-regular graph.
Region targets: Connected planar: $r = m - n + 2$. W_n gives $r = n$. For $n = 6, r = 8$, need $m = 12$; octahedral graph works.
Complement nonplanarity: If $G, \overline{G}$ both planar and $n \geq 11$, then $\binom{n}{2} = m + \bar{m} \leq 6n - 12$, false.
Standard nonplanar: K_5 : $n = 5, m = 10$, nonplanar since $10 > 3(5) - 6$. $K_{3,3}$: $n = 6, m = 9$, nonplanar since bipartite planar would require $9 \leq 2(6) - 4 = 8$.
Deleting one edge: $K_5 - e$ has $n = 5, m = 9$ and is planar; $K_{3,3} - e$ has $n = 6, m = 8$ and is planar. Edge bounds become tight but do not by themselves prove planarity.

Prove planar: Give a crossing-free drawing/embedding, or list face boundaries where each edge appears exactly twice and Euler's formula is satisfied. Edge bounds alone cannot prove planar.
Disprove planar: Use bound violation, or find subdivision/minor of K_5 or $K_{3,3}$ (Kuratowski-type). For bipartite graphs, try $m \leq 2n - 4$ first.
Warnings: A bad drawing does not prove nonplanarity; one good drawing proves planarity. Regions sharing only a corner are not adjacent in map coloring.

Trees

Definitions: Tree = connected acyclic graph. Forest = acyclic graph; components are trees. Spanning tree of connected G : tree subgraph with all

vertices.
Equivalences: For finite simple T with n vertices, equivalent: connected+acyclic; connected and $m = n - 1$; acyclic and $m = n - 1$; unique simple path between every pair; connected and every edge is a bridge.
Forest formula: Forest with n vertices and c components has $m = n - c$. Tree has $m = n - 1$. Every nontrivial tree has at least two leaves; removing a leaf leaves a smaller tree.
High-degree tree forms: Deg $n - 1$ vertex $\Rightarrow$ star $K_{1,n-1}$. Deg $n - 2$ vertex $\Rightarrow$ star with one pendant edge subdivided once.
Bridge deletion proof: Tree $\Rightarrow$ deleting any edge disconnects; otherwise endpoints remain connected and form a cycle. Conversely, connected and every edge bridge $\Rightarrow$ tree.
Rooted tree terms: Root, parent, child, siblings, ancestors, descendants, leaf, internal vertex, level, height.
Full binary tree: Every internal vertex has exactly two children. If I internal vertices, leaves $L = I + 1$, total vertices $2I + 1$, edges $2I$.
Perfect binary tree: Height h : level k has 2^k vertices, leaves 2^h , total vertices $2^{h+1} - 1$.
 $K_{p,q}$ tree criterion: $K_{p,q}$ is a tree iff $p = 1$ or $q = 1$, since $pq = p + q - 1 \iff (p - 1)(q - 1) = 0$.
MST / Shortest Paths
MST: Weighted connected graph $G = (V, E, W)$. A spanning tree has all vertices and $n - 1$ edges. MST minimizes total edge weight.
Cut property: A unique lightest edge crossing a cut belongs to every MST. Such an edge is safe.
Cycle property: A unique heaviest edge on a cycle belongs to no MST. Distinct edge weights imply unique MST; equal weights may give several MSTs.
Kruskal: Sort all edges by increasing weight; add next edge iff it does not create a cycle; stop after $n - 1$ accepted edges. Accepted edges always form a forest.
Prim: Start with vertex set $S = \{s\}$; repeatedly add cheapest edge crossing $S, V \setminus S$ and add the new vertex. Selected edges form one growing tree.
Dijkstra: Requires nonnegative edge lengths. Initialize $d(s) = 0, d(v) = \infty$. Repeatedly settle unsettled vertex u with smallest $d(u)$; relax $d(v) \leftarrow \min\{d(v), d(u) + \ell(u, v)\}$. Settled distances are final. Store predecessor $\pi(v)$ to reconstruct path. Stop after target is settled.
MST vs SPT: MST minimizes total connection cost; shortest-path tree minimizes source distances. They may differ.
MBST: Minimum bottleneck spanning tree minimizes largest selected edge. Every MST is MBST, not conversely. Bottleneck $\leq B$ iff subgraph of edges of weight $\leq B$ is connected.
Prim/Kruskal verification: Prim: chosen edge must be minimum across current cut $(S, V \setminus S)$. Kruskal: rejected edge closes a cycle; accepted edges join different components.
Weighted K_n trick: For K_n with $w(\{u, v\}) = u + v$, MST is star $K_{1,n-1}$ centred at 1; total $\sum_{i=2}^n (1 + i) = \frac{(n-1)(n+4)}{2}$.
Counting traps: Ordered vs unordered; distinct vs identical; labelled vs unlabelled boxes; repetition allowed? cases disjoint? complement universe fixed? symmetry counted equally before division? upper bounds included? leading zeros allowed?
Identity traps: Define counted set before algebra. For sums, identify what the summation index classifies. For coefficient extraction, match exponents; do not expand unless necessary.
Recurrence traps: Define state and initial conditions; state valid range; repeated roots need powers of n ; non-homogeneous needs particular solution; resonance needs extra n^m ; check first terms.
Graph traps: Read adjacency, not drawing. Degree sequence is only necessary. Euler criteria need connectedness. Hamilton is not Euler. Planar edge bounds are necessary only. Dijkstra requires nonnegative weights. MST is not shortest path.
One-line triggers: “At least one” $\rightarrow$ complement/IE. “Guaranteed” $\rightarrow$ PHP. “Prove identity” $\rightarrow$ double count. “No consecutive” $\rightarrow$ last-symbol recurrence/gaps. “All degrees” $\rightarrow$ handshaking. “Use every edge” $\rightarrow$ Euler. “Visit every vertex” $\rightarrow$ Hamilton. “Two colors” $\rightarrow$ bipartite/no odd cycle. “No crossings” $\rightarrow$ Euler formula, planar bounds, Kuratowski. “Minimum total connection” $\rightarrow$ MST. “Shortest route from source” $\rightarrow$ Dijkstra.